Attention-Based Segmentation from Stable Diffusion 1.5

Ablations, interpretation, and cross-dataset generalization

October 2026

1 Purpose and main result

We segment objects using Stable Diffusion 1.5 attention without training a segmentation network. VOC-Sim (490 images) is the development set; COCO-Cap (707 images) is an untuned transfer test. The main result is that **self-attention propagation provides most of the improvement**.

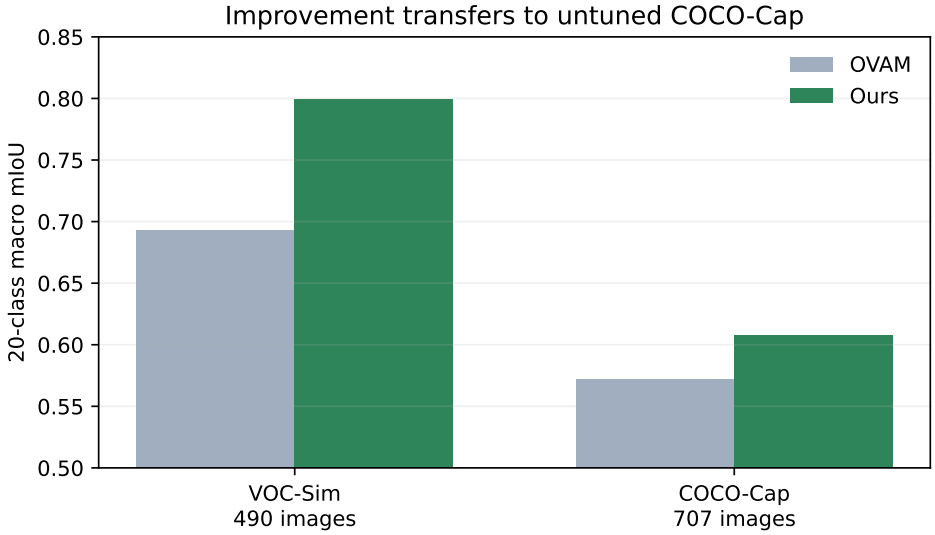


Figure 1: Dataset-level comparison. The COCO-Cap method and all its hyperparameters were frozen from VOC-Sim before evaluation.

2 Experimental protocol

Each released prompt and seed is regenerated for 30 inference steps. Maps are scored against the released mask and then discarded. mIoU is the 20-class macro average of pooled foreground IoU. OVAM is reproduced with its released dataset-specific thresholds and DenseCRF; token optimization is excluded.

Our final configuration uses a top-four-head CA seed from 12 selected layers, the final 64×64 decoder SA module, four propagation passes, SA heads $\{0, 1, 2, 4, 5, 6\}$, denoising calls 15–30, percentile 1–99 normalization, threshold 0.45, and removal of connected components below 1% of the image.

3 Cross-attention ablations

3.1 Denoising time

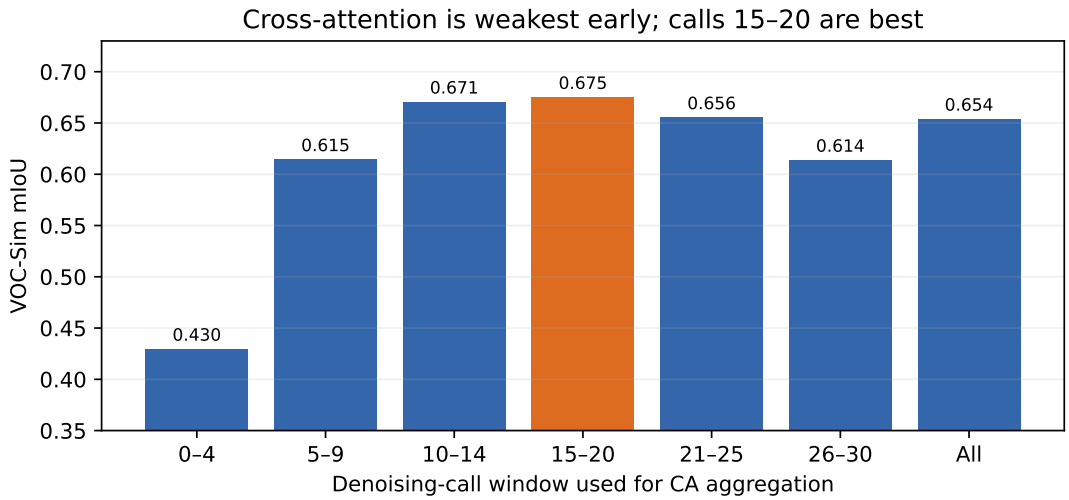


Figure 2: CA-only time-window ablation on all 490 VOC-Sim images. Very early calls are poorly localized; calls 15–20 are strongest in this aggregation.

CA calls	0–4	5–9	10–14	15–20	21–25	26–30	All
mIoU	0.430	0.615	0.671	**0.675**	0.656	0.614	0.654

Table 1: Numeric CA time-window results.

Calls 15–20 are best; the earliest calls are poorly localized.

3.2 Resolution

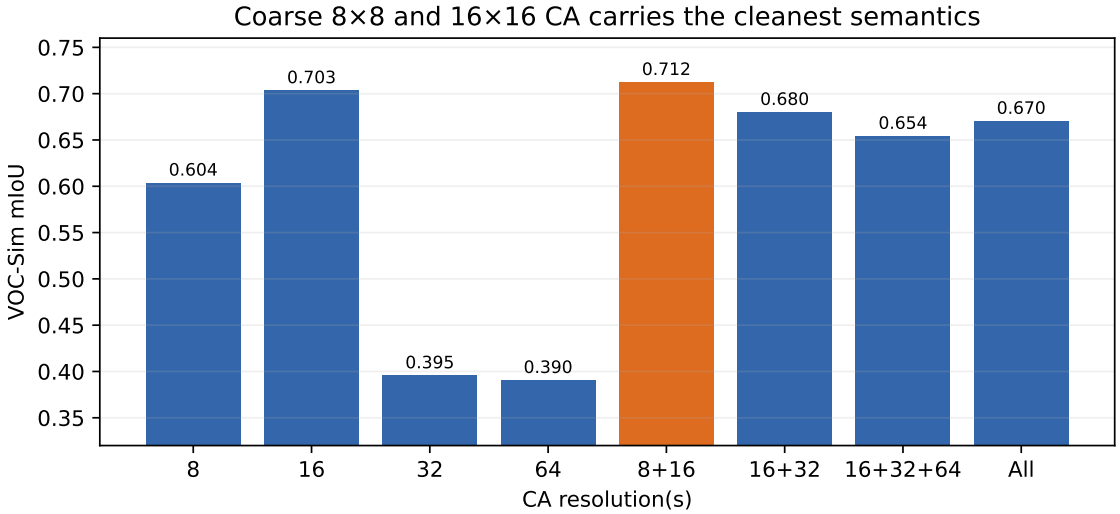


Figure 3: CA resolution ablation. Combining 8×8 and 16×16 maps is best (0.712 mIoU); 32×32 and 64×64 CA alone are weak.

CA resolution	8	16	32	64	8+16	16+32	16+32+64	All
mIoU	0.604	0.703	0.395	0.390	**0.712**	0.680	0.654	0.670

Table 2: Numeric CA resolution results.

Coarse 8×8 and 16×16 CA carries cleaner semantics than the higher-resolution CA maps.

3.3 Individual layers

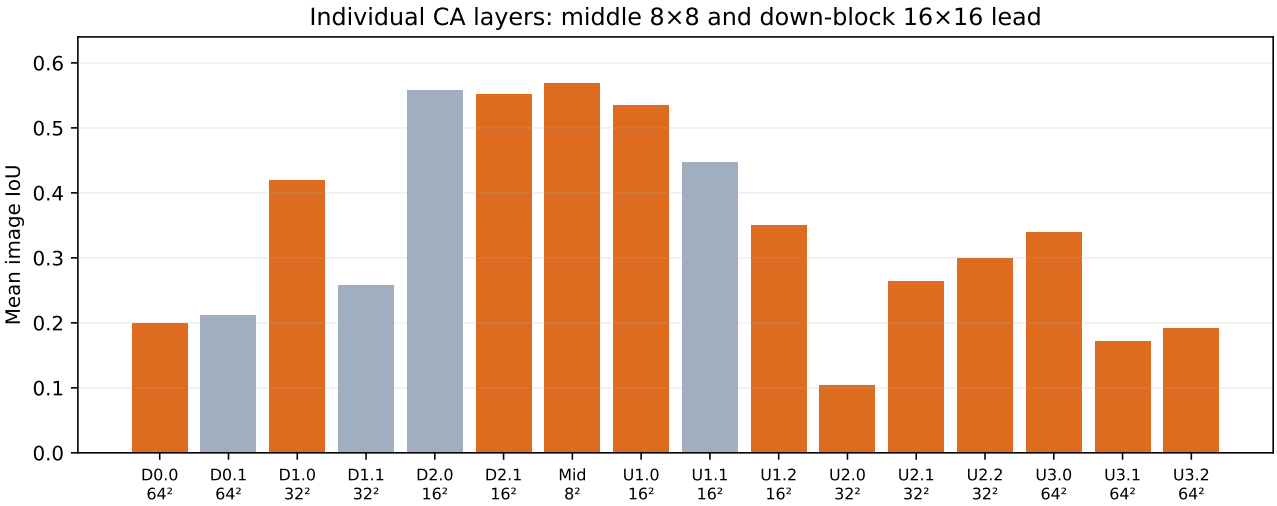


Figure 4: Individual CA layers. Labels give U-Net region and resolution; orange marks the 12-layer subset inherited by the final method. The middle 8×8 block and two down-block 16×16 layers are strongest.

Layer	Res.	IoU	Selected	Layer	Res.	IoU	Selected
D0.0	64	0.200	yes	U1.1	16	0.446	no
D0.1	64	0.211	no	U1.2	16	0.350	yes
D1.0	32	0.420	yes	U2.0	32	0.104	yes
D1.1	32	0.258	no	U2.1	32	0.264	yes
D2.0	16	0.559	no	U2.2	32	0.299	yes
D2.1	16	0.552	yes	U3.0	64	0.340	yes
Mid	8	**0.569**	yes	U3.1	64	0.171	yes
U1.0	16	0.534	yes	U3.2	64	0.192	yes

Table 3: Individual CA layer IoU and membership in the selected 12-layer set.

The middle 8×8 block is strongest individually. Individual rank does not explain the selected subset: for example, D2.0 is strong but excluded.

3.4 What layer and head selection actually contributes

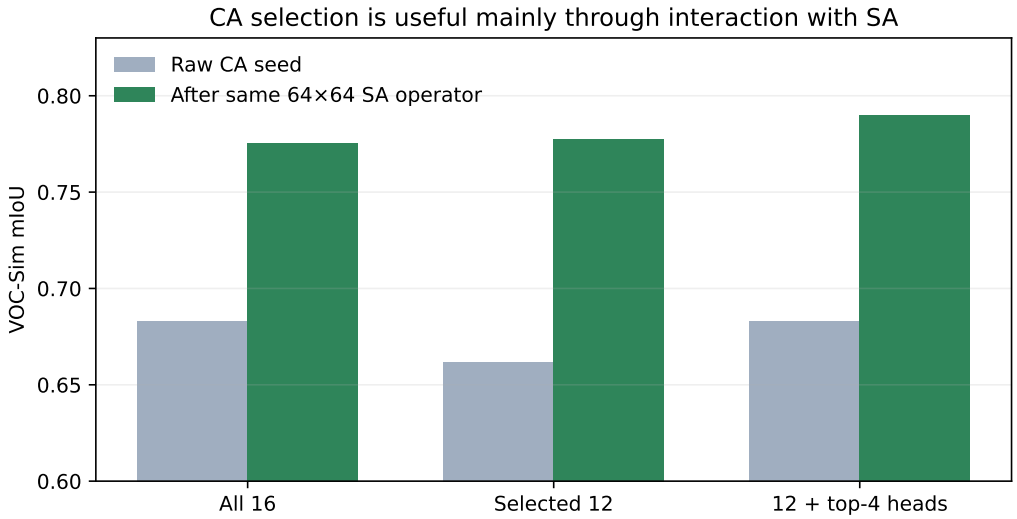


Figure 5: Matched seed comparison before and after the same final-64^2 SA operator (four passes, percentile normalization, threshold 0.45).

CA seed	Raw mIoU	After matched SA	SA gain
All 16 layers/all heads	0.6830	0.7755	+0.0925
Selected 12/all heads	0.6618	0.7777	+0.1159
Selected 12/top-four heads	0.6831	**0.7897**	+0.1066

Table 4: CA selection is not a monotonic improvement path.

Layer selection hurts raw CA. Top-four CA heads help only after SA propagation, which indicates an interaction rather than an independently strong CA choice.

4 Self-attention ablations

4.1 Which SA module?

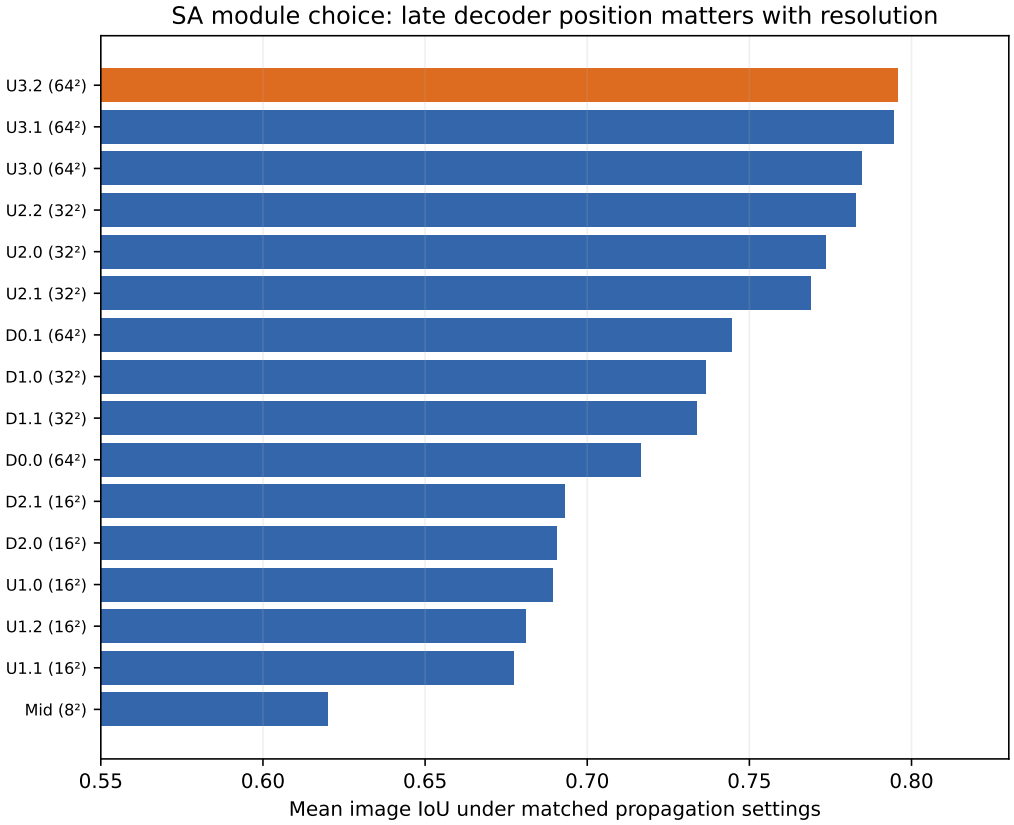


Figure 6: Every SA module under matched seed, propagation, and late-window settings. The final 64×64 upsampling module is best.

SA module	Res.	IoU	SA module	Res.	IoU
D0.0	64	0.717	U1.1	16	0.677
D0.1	64	0.745	U1.2	16	0.681
D1.0	32	0.737	U2.0	32	0.773
D1.1	32	0.734	U2.1	32	0.769
D2.0	16	0.691	U2.2	32	0.783
D2.1	16	0.693	U3.0	64	0.785
Mid	8	0.620	U3.1	64	0.794
U1.0	16	0.689	U3.2	64	**0.796**

Table 5: SA module results under matched settings.

The last 64×64 decoder module is best. Both decoder position and resolution matter; the early 64×64 modules are much weaker.

4.2 How many propagation passes?

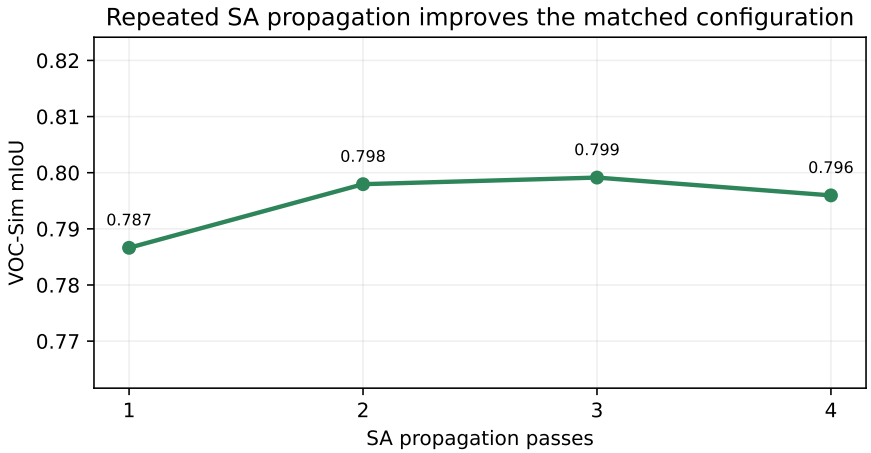


Figure 7: Propagation depth with heads, time window, normalization, and threshold held fixed.

SA passes	1	2	3	4
mIoU	0.7867	0.7979	**0.7991**	0.7957

Table 6: Propagation-depth sweep in the earlier fixed configuration.

Repeated propagation helps, with similar results from two to four passes. In the separate final matched control, one versus four passes scores 0.7799 versus 0.7991.

4.3 Which denoising calls?

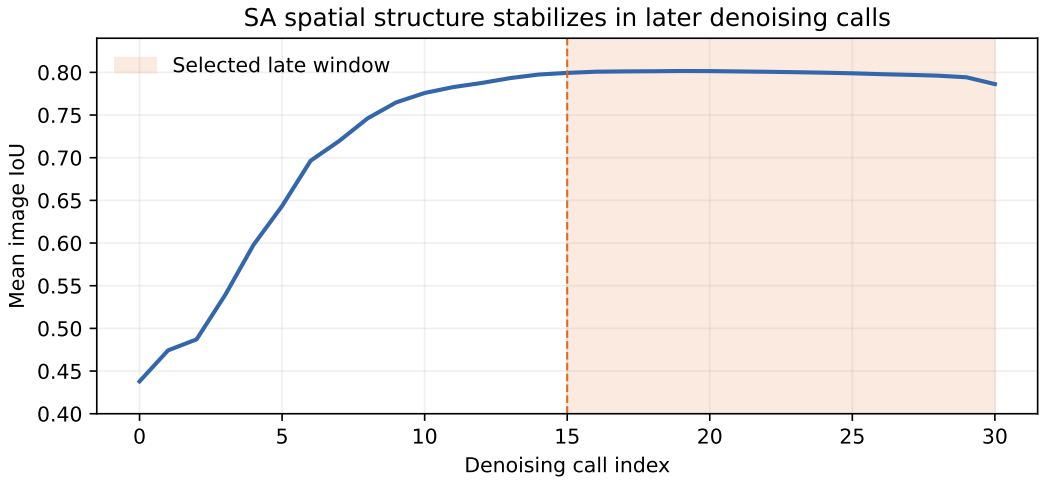


Figure 8: Individual-call quality for the winning SA module. The shaded calls 15–30 form the selected late window.

Call	IoU	Call	IoU	Call	IoU
0	0.438	11	0.783	21	0.801
1	0.474	12	0.788	22	0.801
2	0.487	13	0.793	23	0.800
3	0.539	14	0.797	24	0.800
4	0.598	15	0.799	25	0.799
5	0.643	16	0.801	26	0.798
6	0.696	17	0.801	27	0.797
7	0.720	18	0.801	28	0.796
8	0.746	19	**0.802**	29	0.794
9	0.765	20	0.801	30	0.786
10	0.776				

Table 7: Individual-call IoU for the selected SA module.

Calls 15–30 average 0.799 IoU versus 0.670 for calls 0–14. The matched late-window gain over all calls is 0.0054 mIoU.

4.4 Which SA heads?

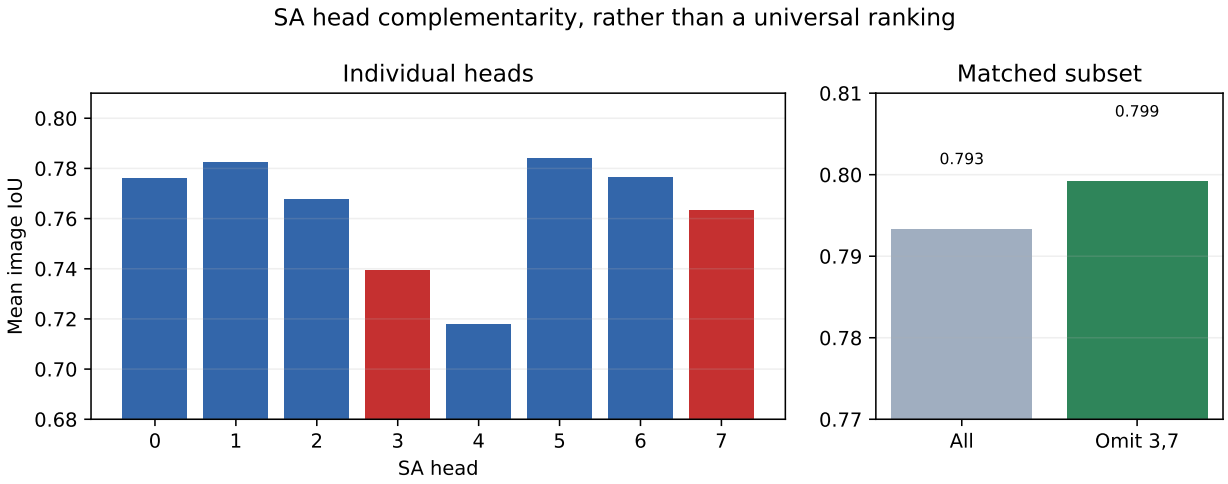


Figure 9: Left: each head alone. Right: the matched aggregate. Removing heads 3 and 7 raises mIoU from 0.7929 to 0.7991.

Head	IoU	Head	IoU	Aggregate	mIoU
0	0.776	4	0.718	All heads	0.7929
1	0.782	5	**0.784**	Omit 3,7	**0.7991**
2	0.768	6	0.776		
3	0.739	7	0.763		

Table 8: Individual SA heads and the matched aggregate comparison.

The result reflects head complementarity: retained head 4 is weak alone, yet omitting heads 3 and 7 improves the aggregate by 0.0062 mIoU.

5 What changes in the final mask?

Measure	Implemented OVAM	Final method	Change
Average precision of continuous map	0.899	0.946	+0.047
ROC AUC	0.947	0.968	+0.021
Boundary-gradient ratio	1.774	3.951	+2.177
Centroid error	0.051	0.037	−0.014
Predicted foreground area	0.467	0.382	−0.085
Ground-truth foreground area	0.370	0.370	—

Table 9: Map diagnostics over 490 VOC-Sim images.

Precision rises from 0.710 to 0.869 with nearly unchanged recall (0.895 to 0.897). The gain mainly comes from suppressing false foreground.

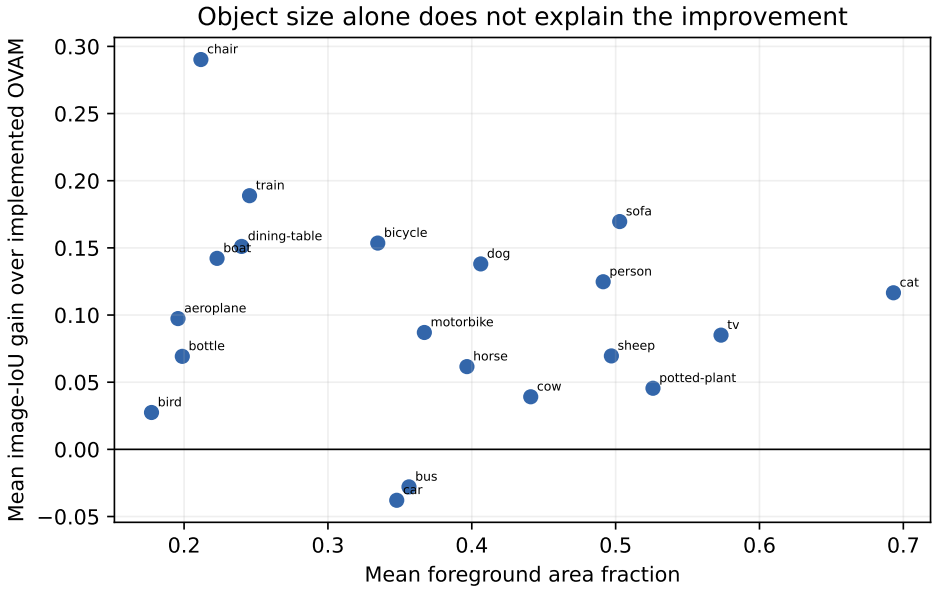


Figure 10: Class-average object area versus gain. Size is not a sufficient explanation; the largest-area quartile gains less, but all quartiles improve.

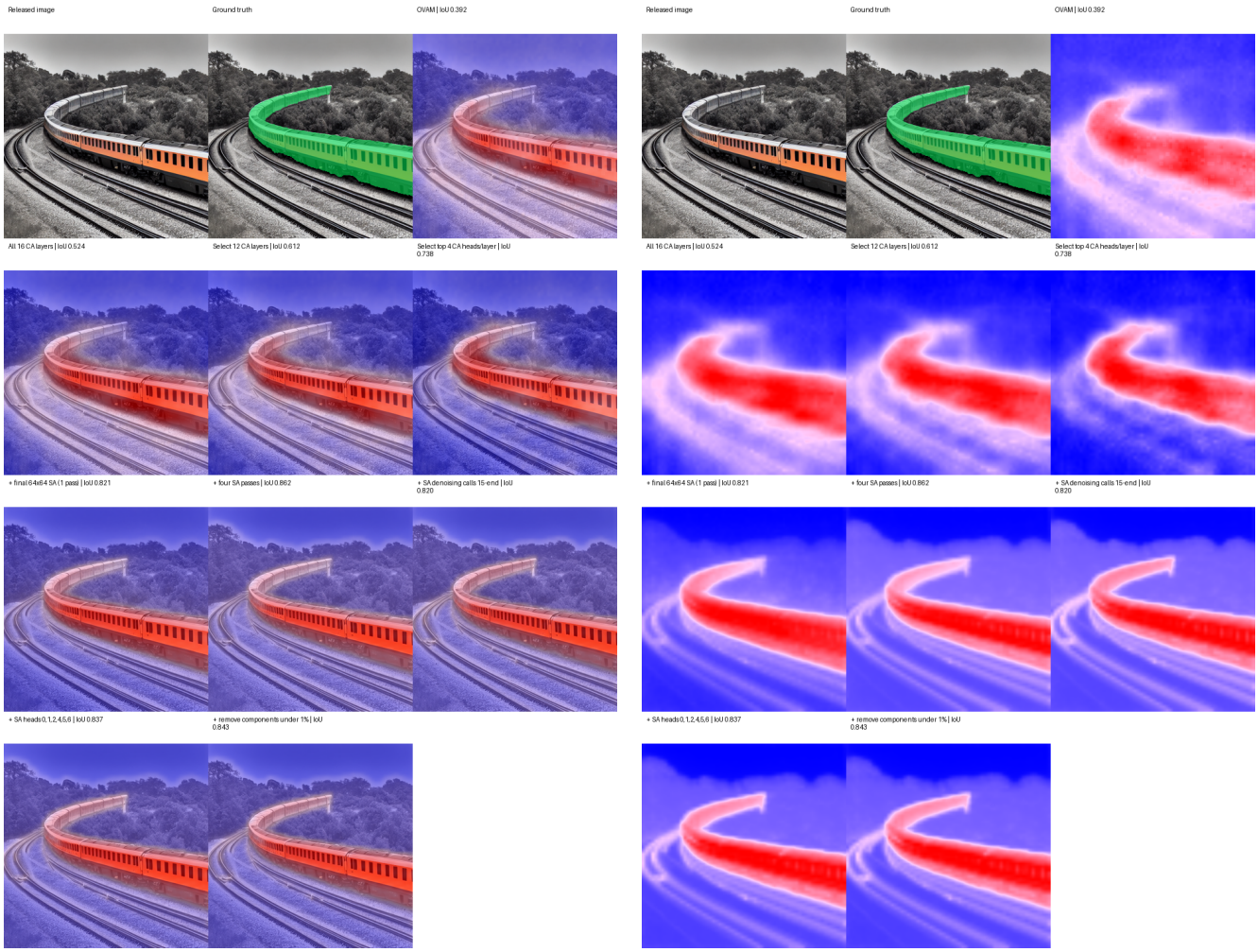
Object-area quartile	Images	Mean image-IoU gain	Win rate
Q1 (smallest)	123	+0.098	72.4%
Q2	122	+0.127	79.5%
Q3	122	+0.131	84.4%
Q4 (largest)	123	+0.044	70.7%

Table 10: Improvement by ground-truth foreground-area quartile.

Every size quartile improves, although gains are smaller for the largest objects.

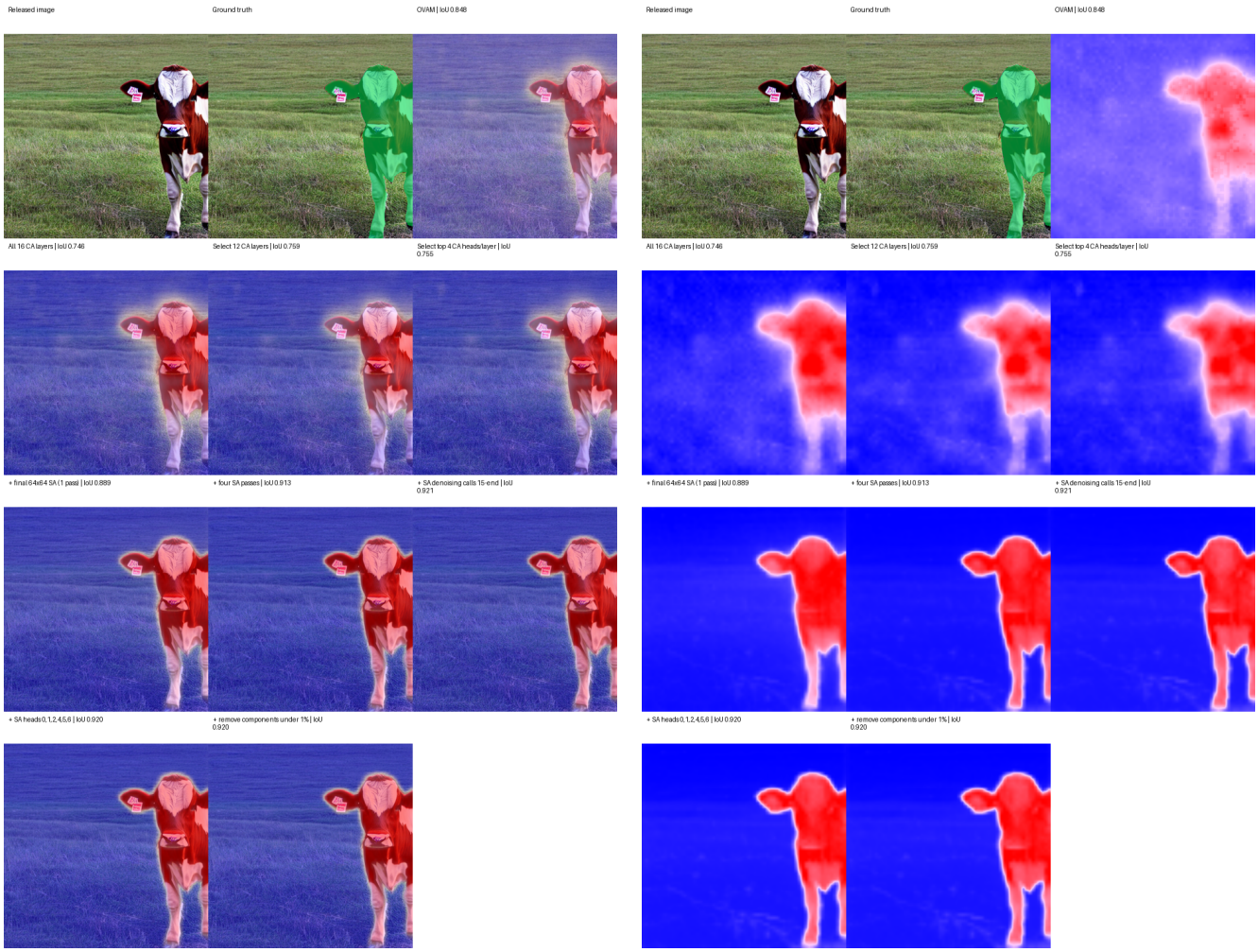
6 Qualitative progression

Each panel shows OVAM, CA choices, SA propagation, late calls, selected heads, and cleanup. Every tile includes image IoU.



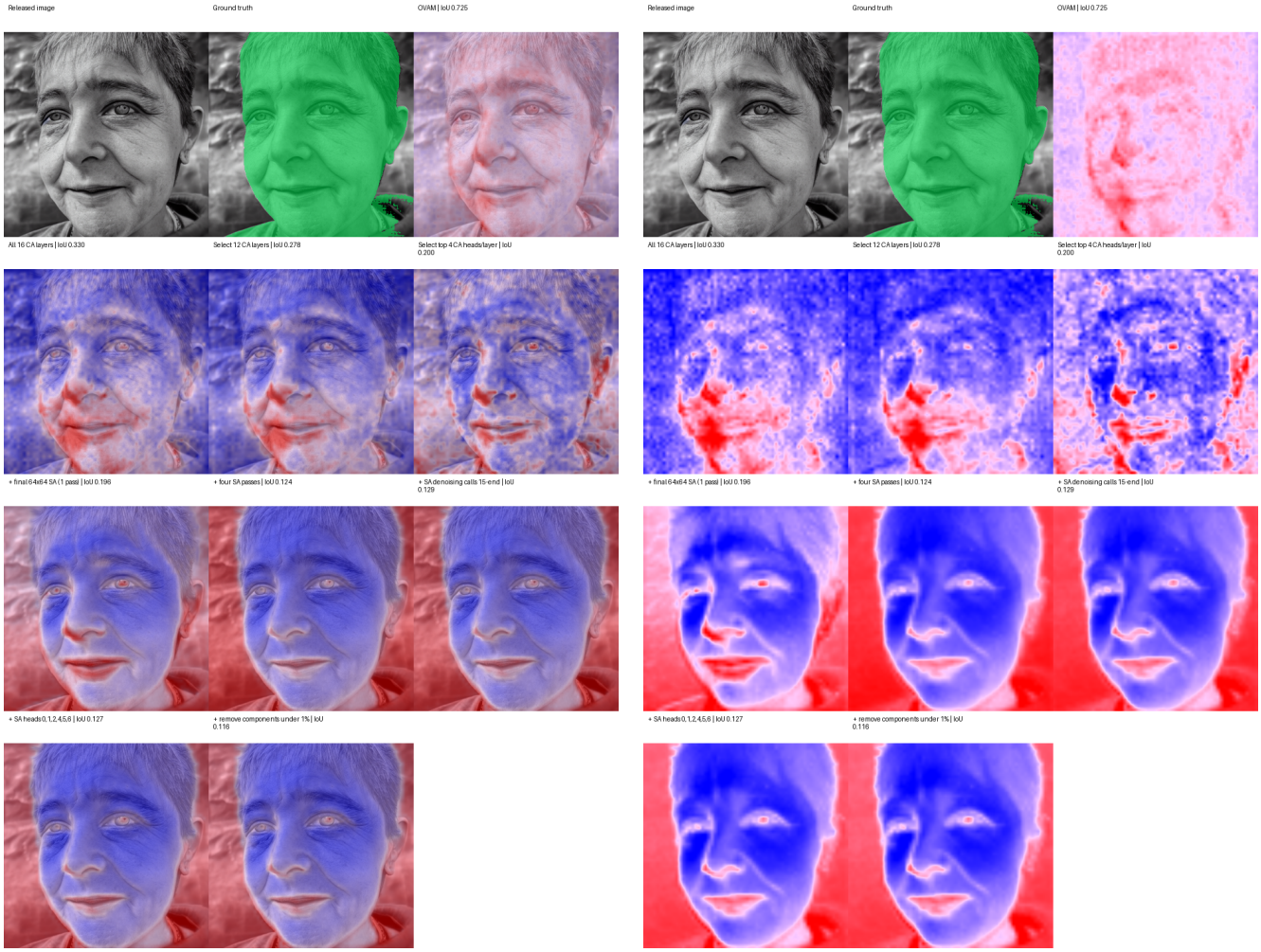
(a) Overlays on the released image. **(b)** Attention and mask maps alone.

Figure 11: Large gain: train, OVAM 0.392 → final 0.843.



(a) Overlays on the released image. **(b)** Attention and mask maps alone.

Figure 12: Representative strong case: cow, OVAM 0.848 → final 0.920.



(a) Overlays on the released image. **(b)** Attention and mask maps alone.

Figure 13: Failure case: person, OVAM 0.725 → final 0.116. The method can overfocus on a subset of a large articulated object.

7 Generalization to COCO-Cap

No choice was tuned on COCO-Cap. The frozen method improves 12 of 20 classes.

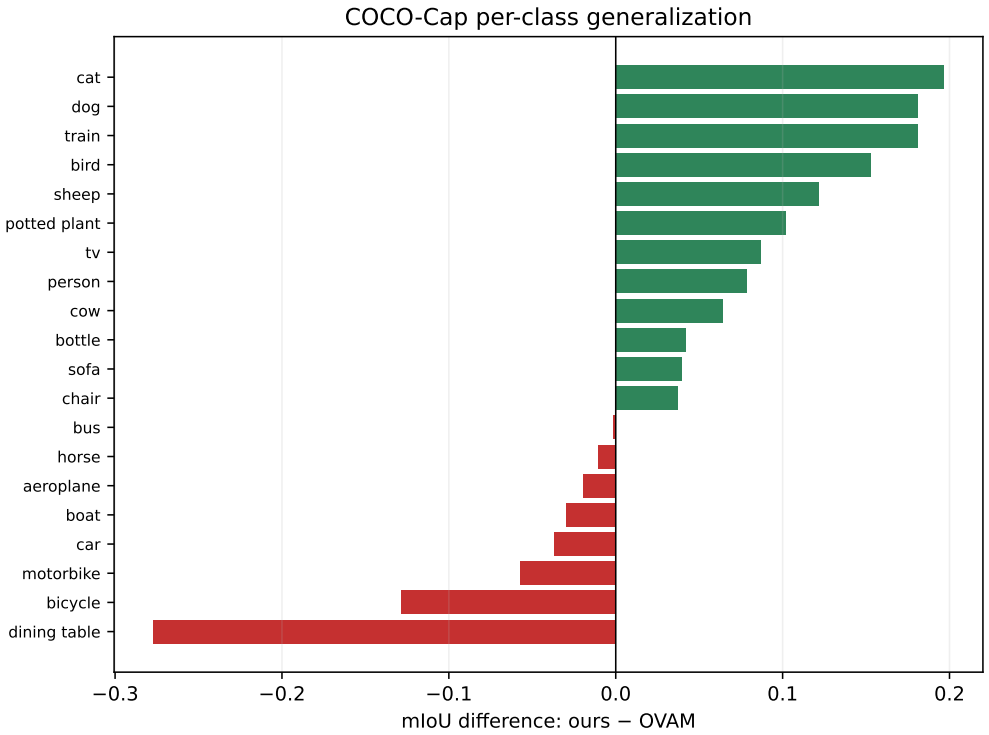


Figure 14: Per-class mIoU change on COCO-Cap. Gains are strong for cat, dog, train, bird, and sheep; dining table and bicycle regress substantially.

Class	OVAM	Ours	Δ	Class	OVAM	Ours	Δ
aeroplane	0.646	0.627	-0.019	dining table	0.896	0.619	-0.277
bicycle	0.639	0.510	-0.129	dog	0.564	0.745	+0.181
bird	0.551	0.704	+0.153	horse	0.716	0.705	-0.010
boat	0.516	0.486	-0.030	motorbike	0.645	0.588	-0.057
bottle	0.326	0.368	+0.042	person	0.194	0.272	+0.078
bus	0.852	0.850	-0.002	potted plant	0.544	0.646	+0.102
car	0.473	0.436	-0.037	sheep	0.739	0.860	+0.122
cat	0.651	0.847	+0.196	sofa	0.367	0.406	+0.040
chair	0.280	0.317	+0.037	train	0.547	0.728	+0.181
cow	0.765	0.829	+0.064	tv	0.528	0.615	+0.087

Table 11: COCO-Cap pooled foreground IoU by class.

The gain persists whether the class word is present (0.5666 → 0.6016) or absent (0.5857 → 0.6170).

8 Final comparison

Dataset	Method	Images	Classes	mIoU	Gain over OVAM
VOC-Sim	Implemented OVAM	490	20	0.6934	—
VOC-Sim	Ours	490	20	**0.7997**	**+0.1064**
COCO-Cap	OVAM	707	20	0.5719	—
COCO-Cap	Ours, frozen from VOC	707	20	**0.6080**	**+0.0362**

Table 12: Primary comparison. VOC-Sim is the development dataset; COCO-Cap is the untuned generalization dataset.

The authors' released VOC-Sim masks score 0.7036. We use our live 0.6934 implementation for the matched comparison because both methods use the same regenerated trajectories.